ASSIGNMENT 1

PAYMENT CALCULATOR FOR A VILLAGE MILK CO-OPERATIVE

Task 5: Read Input and Convert Types

Aim: To read farmer details, convert input into appropriate data types, validate the values and display the results neatly.

Input conversion: Python's `input()` function always returns a string. Numeric text must be converted before performing calculations. The `Decimal` type is used here to support accurate decimal arithmetic for milk payments.

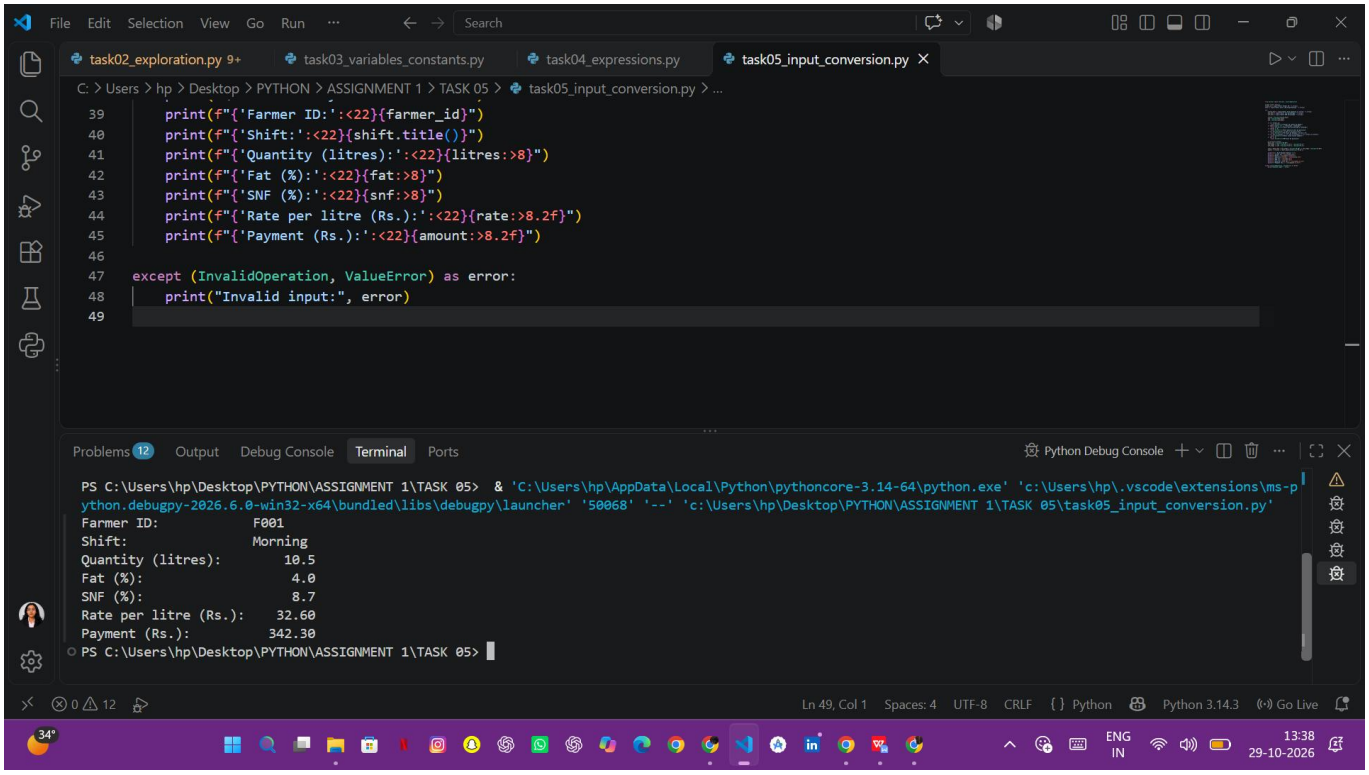
Invalid input handling: The program uses `try` and `except` to handle invalid numeric input. Text such as `eight` and decimal values written as `8,5` are rejected. The program also checks that the farmer ID is not empty, the shift is valid, the quantity is positive, and the fat percentage is within the specified range.

Formatted output: F-strings with field widths align the labels and values. The format specifier `.2f` displays monetary amounts with two decimal places.

Alternative concept: Another approach is to use `float(text)` inside a `try/except ValueError` block. It accepts values such as `8.5` and surrounding spaces, but rejects `eight` and `8,5`. For this payment system, `Decimal` is preferred for monetary calculations.

Complexity: Input conversion and validation take O(1) time per entry for ordinary input lengths and use O(1) auxiliary space.

Conclusion: Input conversion, validation and exception handling help prevent invalid entries and make the payment calculator more reliable.


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task02_exploration.py 9+
task03_variables_constants.py
task04_expressions.py
task05_input_conversion.py
C: > Users > hp > Desktop > PYTHON > ASSIGNMENT 1 > TASK 05 > task05_input_conversion.py > ...
print(f"{'Farmer ID:':<22}{farmer_id}")
print(f"{'Shift:':<22}{shift.title()}")
print(f"{'Quantity (litres):':<22}{litres:>8}")
print(f"{'Fat (%):':<22}{fat:>8}")
print(f"{'SNF (%):':<22}{snf:>8}")
print(f"{'Rate per litre (Rs.):':<22}{rate:>8.2f}")
print(f"{'Payment (Rs.):':<22}{amount:>8.2f}")
except (InvalidOperation, ValueError) as error:
print("Invalid input:", error)
Problems 12 Output Debug Console Terminal Ports
Python Debug Console
PS C:\Users\hp\Desktop\PYTHON\ASSIGNMENT 1\TASK 05> & 'C:\Users\hp\AppData\Local\Python\pythoncore-3.14-64\python.exe' 'c:\Users\hp\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '50068' '--' 'c:\Users\hp\Desktop\PYTHON\ASSIGNMENT 1\TASK 05\task05_input_conversion.py'
Farmer ID: F001
Shift: Morning
Quantity (litres): 10.5
Fat (%): 4.0
SNF (%): 8.7
Rate per litre (Rs.): 32.60
Payment (Rs.): 342.30
PS C:\Users\hp\Desktop\PYTHON\ASSIGNMENT 1\TASK 05>
Ln 49, Col 1 Spaces: 4 UTF-8 CRLF Python Python 3.14.3 Go Live
ENG IN
13:38
29-10-2026